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Peltier-Cooled Electron Cloud Chamber

By grussworks in Design > 3D Design ♥ 1



Our completed Peltier-cooled cloud chamber, designed to make invisible charged particles visible using alcohol vapor and a temperature

differential.

Supplies



Supplies Shown:

TEC2-19006 Thermoelectric Peltier Cooler Module

- This is the main cooling component. When DC power is applied, one side gets cold and the other side gets hot. We use the cold side

to chill the metal plate inside the cloud chamber.

CPU Tower Heatsink with Cooling Fan

- This removes heat from the hot side of the Peltier module. The Peltier needs strong heat removal, or the cold side will not reach a low enough temperature.

AC to DC 12V Switching Power Supply

- This converts wall power into 12V DC power for the Peltier module and fan. It gives the cooling system the steady power it needs to run.

ARCTIC MX-4 Thermal Compound

- This improves heat transfer between the Peltier, cold plate, and heatsink. It fills small air gaps so the cooling stack works more efficiently.

Thermal Paste Spreader Tool

- This is used to spread the thermal compound into a thin, even layer before pressing the parts together. I highly suggest purchasing a spread tool, as the paste is a key component to reaching desired temperatures.

Not shown: Aluminum plate

...There will be some variation in parts and hardware to combine everything. The most important part is to find a plate that is roughly the size of the peltier, as well as, a cpu heat sink the size of the peltier. Ideally, the surface area of the peltier is equal to that of the heat sink and plate. I have included a full parts list with links at the end, however, I suggest finding parts based on your chamber size and desired applications for the device.

Step 1: Setting Up the Environment



Before building the final device, we used two bench power supplies to test the Peltier, fan, heatsink, and aluminum plate separately.

Before we designed the final housing, we first built and tested the cooling system on the table using a plastic cup. This was the setup stage where we made sure the Peltier cooler could actually bring the aluminum plate down to the temperature range needed for the cloud

chamber. This first test allowed us to see the weight each change had on the temperature drop. For example, the heat sink seemed less important than the thermal paste, Although necessary to have a good heat sink, adding a generous amount of thermal paste brought the system from around 0 C to the needed -30 C. So, if the plate is not reaching the low temp, I suggest experimenting with different levels of paste.

We used two bench power supplies during testing. One power supply was used for the Peltier module, and the other was used for the fan. This made testing easier because we could control and monitor the voltage and current going into each part separately.

The main goal of this step was to see if we could reach the desired cold plate temperature before locking everything into the final 3D printed body.

For the cooling stack to work, the parts had to be in direct contact in this order:

Aluminum plate → thermal paste → Peltier cold side → thermal paste → heatsink → fan

The aluminum plate, Peltier, and heatsink all need to be pressed together, having full contact. If there is an air gap between any of these parts, the cooling performance drops significantly. The Peltier can only move heat efficiently if the cold side is touching the plate and the hot side is touching the heatsink.

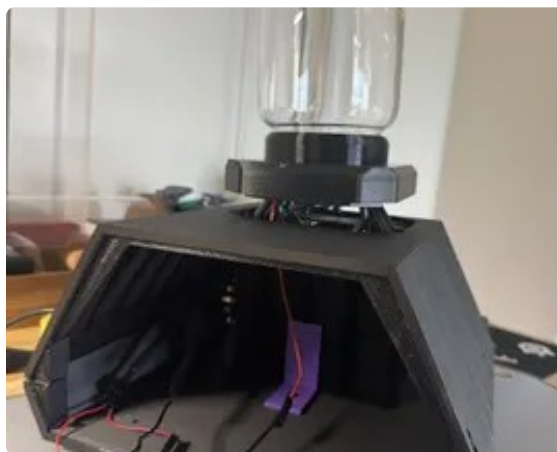
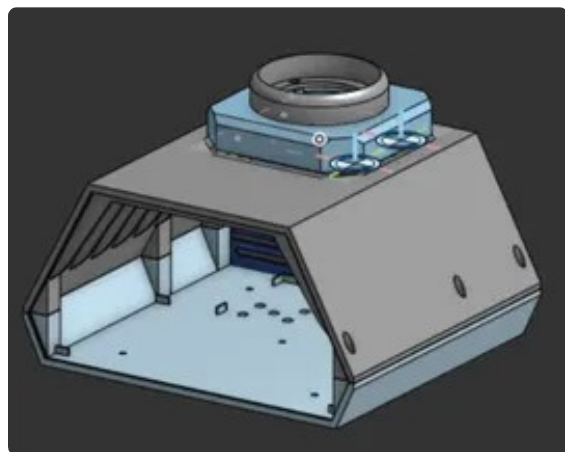
Thermal paste was also very important, as I mentioned above. The paste fills in tiny gaps between the surfaces that look flat but are not perfectly smooth. Better contact means better heat transfer. The more evenly the thermal paste is applied, the better the cooling can be, but only up to a certain limit. A thin, but complete layer is best. Too little paste leaves air gaps, but too much paste can become a thick layer that actually slows heat transfer.

For our test, we made sure:

- The Peltier cold side was touching the aluminum plate.
- The Peltier hot side was touching the heatsink.
- Thermal paste was between each contact surface.
- The fan was running before or at the same time as the Peltier.
- The heatsink had open airflow.
- The plate temperature was measured while the system was running.

This step helped us confirm that the cooling system could reach the temperature range needed before we built the full cloud chamber around it.

Step 2: Designing the 3D Printed Housing



We designed the housing around the cooling stack, the fan, and the glass cover from a Yankee Candle container. We had space for the power supply, however it did not yet get delivered.

After the cooling system worked on the bench, we designed the actual 3D printed housing. The housing had to hold all of the parts in the correct position while still allowing airflow, visibility, and access to the chamber. Luckily the heat sink radiates heat at a relatively low temperature, so no 3D components are at risk of melting. The "hot-side" of the peltier could be too hot, so be carefully placing 3D components in direct contact with the heat side; note: heat side does not radiate far, so the components nearby not in contact are safe.

The housing design had two main sections:

1. The fan and heatsink housing

2. The glass chamber housing for the Yankee Candle cover

The fan housing was designed to hold the CPU fan and heatsink in place without blocking airflow. This was important because the hot side of the Peltier creates a lot of heat. If the fan is blocked, the heatsink gets too hot, and the cold plate will stop cooling properly.

The chamber housing was designed to hold the glass cover from a Yankee Candle container. We used the glass as the transparent chamber because it allowed us to see inside while trapping the alcohol vapor above the cold plate. Glass is superior to plastic in our case, as the alcohol eats away and fits itself in the thin cracks of plastic. Ideally, you do not want alcohol pooling up anywhere.

When designing the housing, leave space for:

- The aluminum cold plate
- The Peltier module
- The heatsink
- The fan
- The Yankee Candle glass cover
- Wires from the Peltier and fan
- Airflow around the heatsink
- A stable base for the chamber
- A seal between the glass and the cold plate area

The most important design requirement was that the housing could not

separate the cooling parts. The aluminum plate, Peltier, and heatsink still needed to be pressed directly together. The 3D print was used to hold the system, not to create extra space between the thermal parts.

Step 3: Constructing the Final System

During construction, we placed the cooling stack inside the housing and made sure the fan, heatsink, Peltier, and plate were aligned. Do not forget to apply thermal paste, as well.

Once the housing was printed, we began constructing the final device. We placed the cooling stack into the housing and made sure the parts were lined up correctly.

The cooling stack was assembled in this order:

Aluminum cold plate on top

Thermal paste

Peltier module

Thermal paste

CPU heatsink

CPU fan

We made sure the fan was not blocked by the housing. The fan needs open space to pull or push air through the heatsink. If the fan is too close to a wall, covered by plastic, or blocked by wires, the heatsink will not remove enough heat.

Wiring the Device

We connected the Peltier and fan to the DC bench supplies so the cooling system could run from a stable power source.

For the final device, the cooling system should be connected to a 12V DC power supply. We used 12V because both the Peltier module and the fan are designed to operate around 12V DC.

The basic wiring layout was:

12V positive from power supply → Peltier positive wire

12V negative from power supply → Peltier negative wire

The fan was also powered from the 12V supply:

12V positive from power supply → fan positive wire

12V negative from power supply → fan negative wire

The Peltier and fan should be wired in parallel, not in series. This lets both components receive the correct voltage.

The Peltier draws most of the current. A TEC2-19006 style Peltier can draw around 4–6 amps depending on the power supply and conditions. The fan draws much less current, usually a fraction of an amp, depending on the fan model.

For our build:

- The Peltier needed about 12V and several amps to cool the plate.
- The fan needed 12V and enough current to spin at full speed.
- The power supply needed to provide enough current for both at the same time.

The fan should always be running when the Peltier is running. If the Peltier is powered without the fan cooling the heatsink, the hot side can overheat, and the cold side will stop being cold.

During assembly, we checked that:

- The wires were connected with the correct polarity and covered to avoid a short.
- The fan turned on properly at needed RPM.
- The Peltier cold side faced the aluminum plate.
- The Peltier hot side faced the heatsink.
- The power supply could provide enough current.
- The housing did not block fan airflow.
- The glass chamber sat securely above the cold plate.

Step 4: Run the Device!



Once the device was assembled, we powered the cooling system and waited for the alcohol vapor to become supersaturated.

After the device was fully constructed, we tested it as a working cloud chamber. The purpose of the device is to create a cold supersaturated alcohol layer near the aluminum plate.

To run the device, we followed this order:

1. Add 99% isopropyl alcohol to the felt or absorbent material and place at top of the chamber.
2. Place the glass cover over the cold plate.
3. Make sure the chamber is sealed as well as possible.
4. Turn on the fan.
5. Turn on the Peltier.
6. Watch the temperature of the aluminum plate.
7. Wait for the chamber to stabilize (10-15 mins); speed up by heating top of chamber.
8. Shine a light across the cold plate from the side.
9. Look for mist (supersaturated state), condensation behavior, and particle trails.

As the Peltier runs, it pulls heat away from the aluminum plate and moves it into the heatsink. The fan then removes that heat from the heatsink. This causes the aluminum plate to become cold.

Inside the glass chamber, the alcohol evaporates from the felt and fills the air with vapor. As the vapor moves downward toward the cold plate, it cools. Near the bottom, the alcohol vapor can become supersaturated, meaning it is ready to condense.

When charged particles pass through this supersaturated layer, they ionize the air. Alcohol droplets then form along that ionized path, creating a visible white droplet trail.

What We Should See

At first, we should not expect trails immediately. The system needs time to cool and stabilize.

A rough timeline is:

- 0–2 minutes: The fan and Peltier begin running. The plate starts cooling.
- 2–5 minutes: The plate temperature drops more noticeably. Some condensation may begin.
- 5–10 minutes: Alcohol vapor begins forming a colder layer near the plate.
- 10–15 minutes: The chamber has the best chance of reaching a stable supersaturated state.
- After stabilization: Thin white trails may appear near the bottom of the chamber.

The exact time depends on the plate temperature, alcohol amount, seal quality, room temperature, and airflow.

To see the trails clearly, lighting is very important. We used a light shining sideways across the bottom of the chamber. Looking through the glass at a slight angle makes the trails easier to see.

The best signs that the chamber is close to working are:

- A faint misty layer near the bottom
- Slight shimmering above the cold plate

- Thin white streaks appearing and disappearing
- No large puddles forming on the plate
- No heavy fog filling the entire chamber

If we only see large droplets, then there may be too much alcohol. If we see no mist at all, then there may not be enough alcohol vapor, the chamber may not be sealed, or the plate may not be cold enough.

When the chamber reaches the right conditions, charged particles can leave visible trails in the alcohol vapor.

When working correctly, the cloud chamber turns invisible particle motion into something we can see directly. This is what makes the device useful as a teaching tool; instead of only explaining radiation as an abstract idea, we can show evidence of particles moving through the chamber.

Step 5: Important Parts List (Detailed)

Item 1: 12V DC Power Supply

Supply Name: Upgraded 12V DC Power Supply 480W 40A AC to DC Converter Switching Power Supply for LED Strip, Motor Pump, CCTV, 3D Printer, Electronic Instruments, Lab Computer

Quantity: 1

Amazon Link: [DC Power Supply](#)

Purpose in Project:

This power supply is used as the main electrical power source for the final cloud chamber device. It converts AC wall power into 12V DC power, which is needed to run the cooling system. The Peltier module requires a strong DC supply because it draws several amps while cooling. The fan also runs from 12V DC, so this supply allows both the Peltier and cooling fan to be powered from one main source.

Project Function:

- Converts wall power to 12V DC
- Powers the Peltier cooling module
- Powers the cooling fan
- Provides enough current capacity for high-load testing and

operation

Item 2: Aluminum Cold Plate

Supply Name: uxcell 6Pcs Aluminum Disc, 2" / 60mm Diameter, 0.08" / 2mm Thickness, 5052 Aluminum Circle Plates Round Sheets for Metal Stamping

Quantity: 1 pack of 6 aluminum discs

Amazon Link: [Cold Plate](#)

Purpose in Project:

The aluminum disc acts as the cold plate at the bottom of the cloud chamber. The Peltier module cools this plate, and the alcohol vapor inside the chamber becomes supersaturated near its surface. This plate protects the peltier from the alcohol.

Project Function:

- Serves as the cold surface inside the chamber
- Transfers heat from the chamber into the Peltier
- Creates the cold region needed for alcohol supersaturation
- Provides the viewing surface where particle trails appear
- Protects Peltier

Item 3: CPU Cooler / Heatsink and Fan

Supply Name: Noctua NH-U9S chromax.Black, 92mm Single-Tower
CPU Cooler, Black

Quantity: 1

Amazon Link: [CPU Cooler](#)

Purpose in Project:

The CPU cooler removes heat from the hot side of the Peltier module. When the Peltier cools one side, the opposite side becomes hot. If this heat is not removed, the cold side will stop cooling effectively.

Project Function:

- Removes heat from the hot side of the Peltier
- Prevents the Peltier from overheating
- Allows the aluminum plate to reach lower temperatures
- Provides active air cooling for the device

Item 4: Thermal Paste

Supply Name: ARCTIC MX-4, 4 g, Premium Performance Thermal Paste for All Processors, CPU, GPU, PC, PS4, Xbox, Very High Thermal Conductivity, Long Durability, Safe Application, Non-Conductive, Non-Capacitive

Quantity: 1 tube, 4 g

Amazon Link: [Thermal Paste](#)

Purpose in Project:

Even if two surfaces look flat, they are not perfectly smooth at a microscopic level. Without thermal paste, tiny pockets of air reduce heat transfer. Since air is a poor conductor, those gaps lower the cooling performance. The thermal paste creates better contact between the parts and improves the overall cooling potential of the system.

Project Function:

- Improves heat transfer between parts
- Reduces air gaps in the cooling stack
- Helps the aluminum plate reach lower temperatures
- Increases the efficiency of the Peltier-heatsink connection

Item 5: 3D Printing Filament

Supply Name: PETG Filament, Roughly 1.25 kg

Quantity: Approximately 1.25 kg

Purpose in Project:

The PETG filament is used to 3D print the housing for the cloud chamber. The housing holds the cooling system, fan, heatsink, aluminum plate, wiring, and glass chamber in the correct layout.

PETG is stronger and more heat-resistant than standard PLA, making it a good choice for a device that contains a heatsink, fan, electrical components, and a cold chamber section. The housing needed to be sturdy enough to hold the parts securely while still allowing airflow around the fan and heatsink.

Project Function:

- Forms the main device housing
- Holds the fan and heatsink in place
- Supports the aluminum cold plate
- Creates a base for the Yankee Candle glass chamber
- Organizes the internal layout of the final build

Total Cost:<\$150, using cheaper fan can bring this to under \$100.

Step 6: CAD Files

These were designed for our specific parts, however, you can easily adapt these files to fit your components instead.